

BORONG XU

Research Assistant at SLAI



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ABOUT ME

Research Assistant at the Shenzhen Loop Area Institute (SLAI) and a graduate of HKUST's Computer Engineering program, ranked 1/98 by major GPA. Passionate about robotics and embodied intelligence, with research experience spanning visual-inertial odometry, SLAM, multi-camera localization, and quadruped control, as well as deep learning and LLM-based agent systems.

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING, FIRST CLASS HONORS

- GPA 3.84/4.3
- Major GPA 4.03/4.3 (Ranked 1/98 in program)

Hong Kong SAR

Graduated Jun 2026

University of New South Wales (UNSW Sydney)

VISITING STUDENT (EXCHANGE), COMPUTER SCIENCE AND ENGINEERING

Sydney, Australia

Feb 2025 - May 2025

High School Affiliated to Renmin University of China (RDFZ)

HIGH SCHOOL DIPLOMA(GAOKAO)

- FIRST Robotics Club: Chief Engineer (2021), Alumni Mentor (2022-2024)

Beijing, China

Graduated Jun 2022

EXPERIENCE

Shenzhen Loop Area Institute (SLAI)

RESEARCH ASSISTANT

- Research on robotics and embodied intelligence.

Shenzhen, China

Jun 2026 - Present

PROJECT EXPERIENCE

LEGO Assembly Understanding

ONGOING RESEARCH PROJECT, JOINTLY SUPERVISED BY HKUST RFL AND SLAI

- Research on the automatic understanding of LEGO assembly, combining computer vision and planning to recover step-by-step assembly structure from visual instructions and 3D models.

Supervised by Prof. Ziqi Wang

Mar 2026 - Present

ELEC5660 Introduction to Aerial Robotics

COURSE PROJECT, HKUST

- Developed aerial robot algorithms from scratch, including visual-inertial odometry, Extended Kalman Filter (EKF) state estimation, and autonomous path planning within a simulation environment, enabling robust UAV localization, mapping, and navigation.

Supervised by Prof. Shaojie Shen

Feb 2026 - May 2026

Quadruped Robot Control

RESEARCH PRACTICUM, UNSW SYDNEY

- Built and programmed a Unitree GO2 quadruped robot to perform complex tasks, integrating depth camera for accurate object localization (localization error < 10cm) and LiDAR for SLAM.

Supervised by Prof. Maurice Pagnucco and Prof. Yang Song

Feb 2025 - Aug 2025

Underwater Localization System

FINAL YEAR PROJECT, HKUST

- Developed a Multi-Camera Vision-Based Localization System for Tracking Multiple Autonomous Underwater Vehicles (AUVs). Reduced localization error in complex underwater lighting scenarios, improving positioning accuracy to within 10cm in a 1m×1m range.

Supervised by Prof. Huan Yin and Prof. Fumin Zhang

Jun 2025 - May 2026

HONORS

2022-2026 **Dean's List**, 5 times

HKUST

2023-2026 **Continuing Students Scholarship**, 10,000 HKD/year

HKUST

2023 **Bronze Medal**, Innovation Competition: Our Livable City, We Engineer

HKIE

2022 **Entry Scholarship**, 70,000 HKD

HKUST



SKILLS

- **C/C++**: Eigen, PCL, STL, STM32, ESP32
- **Python**: PyTorch, OpenCV, Numpy
- **ROS/ROS2**: Navigation2, Gazebo, Slam Toolbox, TF
- **Robotics**: SLAM, VIO, EKF, ICP, Path Planning, Control
- **Deep Learning**: YOLO, ViT, RNN, GAN, VAE, Diffusion, LLM
- **Machine Learning**: GLM, SVM, PCA, XGBoost, HMM
- **Hardware**: Circuit Design, Soldering, CAD, 3D Printing

LANGUAGES

English • Fluent (IELTS 7.5)
Mandarin • Mother tongue